Developer AI — Full Setup Guide from Scratch

Prerequisites

Before starting, ensure the following are installed on the system:

1. **Node.js** (v18, v20, or v22 LTS recommended) — nodejs.org
2. **Python** (v3.10, v3.11, or v3.12 recommended) — python.org (check the box *"Add python.exe to PATH"* during installation)
3. **Git** — git-scm.com
4. A **Gemini API Key** — Free from Google AI Studio

Step 0: Clone the Repository

Open Command Prompt (cmd) or PowerShell:

```
cmd
git clone <YOUR_GITHUB_REPO_URL>
cd dizi_developerai
```

PART 1: Backend Setup (FastAPI + AI Service)

Open a terminal window and navigate to the backend folder:

```
cmd
cd backend
```

1. Create a Python Virtual Environment

```
cmd
python -m venv .venv
```

2. Activate the Virtual Environment

- **On Windows (cmd):**

  ```
  cmd
  .venv\Scripts\activate
  ```

- **On Windows (PowerShell):**

  ```
  powershell
  .\.venv\Scripts\Activate.ps1
  ```

- **On Mac/Linux:**

  ```
  bash
  source .venv/bin/activate
  ```

(You will see (.venv) appear in front of the command prompt)

3. Install Backend Dependencies

cmd
pip install -r requirements.txt

(This installs FastAPI, Uvicorn, Google GenAI SDK, PyTorch, Transformers, and Accelerate)

4. Create the .env Secret File

Because .env is ignored by Git for security, the user must create it manually.

Create a file named .env inside the backend folder and add:

env
GEMINI_API_KEY=your_actual_gemini_api_key_here

Note: The server will automatically create sessions.json and daily_usage.json on first launch, so nothing else needs to be manually created.

5. Start the Backend Server

cmd
python -m uvicorn main:app --reload --host 127.0.0.1 --port 8000

You should see:

text
Requirement classifier loaded on cpu.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Application startup complete.

✅ **Backend is running at:** http://127.0.0.1:8000

🅰 **PART 2: Frontend Setup (Angular Dashboard)**

Open a **SECOND / NEW** terminal window and navigate to the project root directory:

cmd
cd dizi_developerai

1. Install Node Dependencies

Run with --legacy-peer-deps to avoid dependency resolution conflicts:

cmd
npm install --legacy-peer-deps

2. Start the Frontend Server

cmd
npm start

(or npx ng serve)

Once compilation finishes, you will see:

text
Application bundle generation complete.
➜ Local: http://localhost:4200/

✅ **Frontend is running at:** http://localhost:4200

🎯 Step 3: Access the Application

1. Open your browser and go to:

   http://localhost:4200

2. Click **Developer AI** in the sidebar.
3. Start asking coding questions. Markdown headings, code blocks, bullet points, and conversation sessions will work automatically!

⚡ Quick Summary of Commands

Server	Terminal Directory	Commands
Backend	dizi_developerai/backend	python -m venv .venv .venv\Scripts\activate pip install -r requirements.txt *(create .env with API key)* python -m uvicorn main:app --reload --host 127.0.0.1 --port 8000
Frontend	dizi_developerai	npm install --legacy-peer-deps npm start